Hate Speech Detection in Social Media for the Kurdish Language

Ari M. Saeed[1](B), Aso N. Ismael[1], Danya L. Rasul[1], Rayan S. Majeed[1], and Tarik A. Rashid[2]

[1] Department of Computer Science, College of Science, University of Halabja, Halabja, Iraq
Ari.said@uoh.edu.iq

[2] Department of Computer Science and Engineering, University of Kurdistan Hewler, Erbil, Iraq
tarik.ahmed@ukh.edu.krd

Abstract. With the rapid growth of technology over the world, especially, on the internet, people enormously use social media freely to express their ideologies. Sometimes the freedom of media is caught up and the rights of others are beaten down by hate speech. Moreover, social media is an easy and vague way to desecrate people, groups, and parties since there is no any way to recognize anonymous users over social media. Testing human speech is common for English, Arabic, and Turkish languages while there is no attempt for the Kurdish language. For that reason, the Kurdish hate speech dataset is collected from comments on the Facebook application as an effort for detecting hate speech and removing them. The raw dataset consists of 6882 comments which are divided into hate and hot hate classes. Support Vector Machine (SVM), Decision Tree (DT), and Naïve Bays (NB) algorithms are implemented and compared. Based on the results, the SVM is found most excellent with the F1 measure being 0.687.

Keywords: Kurdish hate speech detection · Machine learning algorithms · Text classification

1 Introduction

Nowadays, social media is popular media for communicating with people around the world. With increasing the number of users who share their opinion and exchange their idea, the volume of information is increased massively as well. Facebook is an application that is used as a social media in almost all countries and regions. The Kurdistan region in Iraq is one of them in which people utilize Facebook freely. Kurdish people use the Kurdish language for communication through Latin and Arabic scripts for expressing and posting over Facebook. Moreover, there is not any filtration for publishing the posts. Sometimes people use hate and offensive speeches toward others. There is no doubt, that expressing online hate speech toward opposition has a massive impact by comparing with offline hate speech since more people can see it [1–3]. Based on that fact, users, groups, and parties disturb others by desecrating religion, instigating racism, devaluating freedom rights, culture, and so on. When there is not any filtration for blocking a hate

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post in the Kurdish language by internet providers, people can publish hate speeches without caring to filter and banding their posts. Fortunately, detecting hate speech in international languages, such as English is improved and progressed while there is no attempt for the Kurdish language [4]. For those reasons, detecting hate speech for a national language, such as Kurdish has become a necessary and urgent case. In this study, a new model is created for detecting hate speech in the Kurdish language as a new effort by implementing algorithms and techniques in machine learning and data mining.

2 Motivation and Related Work

With publishing a large volume of hate speeches over social media. A new challenge in data mining techniques is raised for detecting those speeches. Hate speech detection has become a new area that researchers work on it due to its negative effect on socials life. The number of researchers that worked in this domain are few who used machine learning algorithms for the detection of hate speech [1]. For that reason, this dataset is collected from comments and called a Kurdish Hate Speech Dataset (KHSD). KHSD is available for use in some fields of data mining techniques, such as (sentiment analysis, text summarization, and text to speech). It can be utilized by other languages, such as Arabic, Persian, and Urdu since the Kurdish language uses, particularly the same alphabet. Moreover, it can be used for automatic systems in artificial intelligence (AI), such as AI assistants, automated content generation, and automated hate speech detection [4, 5]. Indonesian hate speech text was detected by using the skip-gram model and the word2vec method with Text CNN was implemented. Moreover, the imbalanced dataset was solved by using class weight, random undersampling, and oversampling. The results showed that the F-score is 93.70% [6].

SVM and Long Short Term Memory (LSTM) were utilized for detecting Italian hate speech text on Facebook applications. The ten-fold cross-validation technique was used for partitioning the dataset for training and testing data. The results showed that the effectiveness of automatic hate speech detection was more than the manual classification by using those two algorithms. The accuracy of SVM was 64.61%, which was higher than LTSM, which was 60.50. [7] compares five different text datasets for detecting racist and sexist hate speech. Moreover, different classical data mining algorithms were implemented, such as SVM, GBDT, NB, and LR while CNN with LTSM is used as deep learning classifiers. In this paper, the word-embedding, TF-IDF n-gram, and word gram are utilized [5].

3 Experimental Framework

In this section, a series of steps are implemented for generating the right model data mining fields. The steps are Data collection, Data Description, and Labeling, Preprocessing, Classification as shown in Fig. 1:

Start

Data Collection

Data Description and Labeling

Preprocessing

Classification

End

Fig. 1. Experimental framework

3.1 Data Collection

Data collection is the first step in machine learning to predict the right model and it is important to collect data from valid sources. Different challenges have been tackled in hate speech data collection on social media for some languages, such as English, Italic, Arabic, and so on. For collecting the comments of the dataset on the Facebook application, there are different types of tools that have been utilized. Facepager tool that is applied in this project for collecting comments. The comments on this dataset are collected from different TV channels, athletes, religions, politicians, doctors, ministries, and some specific pages. After installing the tool, the first step is to log into the facepager tool, then, create a new database and add a new node to insert the specified Facebook link for getting the comments. The second step is clicking on the fetch data button to get data and it can increase the page size to get more data. The third step is to export fetching data into a CSV file [8].

3.2 Data Description and Labelling

The Kurdish language is spoken by 40 million people with various dialects. The Kurdish language is not unified since the Kurdish of Syria and Turkey use the Kurmanji dialect with Latin script whilst the Sorani dialect is used in Iran and Iraq with Arabic script. In this project, the Sorani dialect is used to collect the dataset. The dataset consists of 6882 comments of users on the Facebook application. On social media, there are different
types of hate speech is concerned based on categories. In this work, hate and hot hate speech are extracted from eight classes (Politic, Religion, Sport, Terrorism, Education, Race, Freedom, and Sex) [9]. The percentage and number of gathering comments are different for each class as shown in Table 1.

Table 1. Numbers and percentage ratio of eight classes

Class	Number of comments	Hate speech's numbers	Not hate speech's number	Percentage
Politic	1081	431	650	15.7%
Religion	842	415	427	12.2%
Sport	978	406	572	14.2%
Terrorism	718	459	259	10.4%
Education	760	272	488	11%
Race	862	683	179	12.5%
Freedom	693	368	325	10%

Sex	948	632	316	13.7%
Total	6882	3666	3216	100%

After reading the 6882 comments accurately one by one, then, the process of labeling comments is started by four annotators to annotate the comments manually. For annota-tors, it is not easy to predict users' expressions due to the comments being enigmatic and it is a big challenge to classify a comment to hate or not. In this project, the comments are classified to hate or not hate based on the meaning of sentences [5, 13].

3.3 Data Preprocessing

In this subsection, a series of preprocessing steps are applied to the dataset to be ready for use in data mining fields. The preprocessing techniques are Tokenization, Normalization, Removing Stop words, and Stemming.

a- b- Tokenization: is the process of splitting the comments for words by removing spaces.

Normalization: is used to normalize the comments by (Removing Special Charac-ters, Punctuation, Numbers, URLs, hashtags & user mentions if it is English, English letter, words, sentences, Arabic letter, Sentences that has no meaning) and replacing Arabic characters that are used instead of Kurdish characters since different scripts among users over social media are used without knowing its Kurdish script or Arabic script as shown in Table 2.

The comments are read one by one, if the comments don't have any meaning and are corrupted, then they are removed. For that purpose, the number of comments after preprocessing steps is decreased from 6882 to 6607 [9–11]. On the other hand,

Table 2. Replacing Arabic characters with Kurdish characters

NOذ	abiharactersه ۆ بـ	Example	Replaced with Kurdish Characters	Example
.1	ي		ی	
	ة ة	يـ3	.	
	ؤ ؤ	بـ	ۆ	
.4	at the end ئ	نوئ	ێ	ێ نو
.5	U+0647) at the end(ه	باره	)U+06D5(ە	ە بار
.6	rarely used in Kurdish ك	جاك	ک	جاک
	)0647+U (06+0(ەbe) (U+06BE(ھ 7.			

abiharacters Example Replaced with Kurdish Characters Example

1. ی ي

3.ؤ ؤ بـ ۆ ة ة يـ

4. ئ at the end ێ نو ێ نوئ

5. ه)U+0647) at the end ه باره)U+06D5(ە بار

6. ك rarely used in Kurdish جاك ک جاک

)0647+U (06+0(ھbe) (U+06BE(7. ھ

stop words are removed to decrease noise and high dimensionality of feature space [4]. Moreover, Stemming is implemented by removing prefixes and suffixes of words to get the stem (root) of words [9].

3.4 Classification

One of the most important points to be discussed is the balanced dataset. In this study, the KHSD data set is 6882 comments that are divided into two categories hate and note hate. It can be said that the KHSD is almost balanced since the number of hate is 3669 while not hate is 3084. Another important factor is using algorithms, such as SVM, NB, and CNN. Cross-validation is used as a technique for dividing the dataset for training and testing sets. In this experiment, 9-fold is used as training for predicting the models, and 1-fold is used as testing to predict the labels. For evaluating the classifiers, four popular criteria are used for detecting hate speech in text. The following equations are used as shown below [12, 13].

$$\text{Precision}=\frac{TP}{TP+FP} \quad (1)$$

$$\text{Recall}=\frac{TP}{TP+FN} \quad (2)$$

$$\text{F1}=2*\frac{\text{Recall}*\text{Precision}}{\text{Recall}+\text{Precision}} \quad (3)$$

$$\text{Accuracy}=\frac{TP+TN}{TP+FP+TN+FN} \quad (4)$$

Three different classification algorithms are implemented on the KHSD dataset. The performance of each algorithm is different based on various tests. Six different tests are applied as shown below:

1. Raw Sentences (RS) are implemented without any preprocessing technique.
2. Raw Sentences are Normalized (RSN).
3. Raw Sentences are Normalized and Stop Words are removed (RSNSW).
4. Raw Sentences are Normalized and Stemmed (RSNS).
5. Raw Sentences are Normalized and Stop Words are removed and then stemmed (RSNSWS).
6. Raw Sentences are Normalized and Stemmed then Stop Words are removed (RSNSSW).

The F1 measure is used for evaluating the classification methods as the average of precision and recall as shown in Table 3.

Table 3. F1 measure for SVM, C4.5, and NV

	RS	RSN	RSNSW	RSNS	RSNSWS	RSNSSW
SVM	0.644	0.643	0.686	0.650	0.687	0.651

DT (C4.5)	0.560	0.584	0.548	0.584	0.583	0.566
NB	0.560	0.541	0.506	0.558	0.548	0.500

As shown in Table 3 the performance of the F1 measure is increased and decreased gradually based on the algorithms and tests. In SVM the F1 measure is 0.644 for RS while RSNSWS is 0.687, which means the F1 measure is increased slightly by 0.043. In DT the value of RSNS and RSN is 0.584 while in RSNSW is 0.548, which means the difference is 0.036. Moreover, RS decreased from 0.560 to 0.548 in RSNSW. In NV the F1 is 0.560 in RS while it is decreased in all other tests. In Fig. 2, the results of the experiment are different, the SVM is the best classifier for detecting hate speeches for all tests by comparing with DT and NV. The F1 for DT and NV are the same for RS while for all other tests DT is better than NV.

Number of Features
1660
1640
1620
1600
1580
1560
1540
1520
1500
1480
1460
Six Experiments
RS RSN RSNSW RSNS RSNSWS RSNSSW

Fig. 2. F1 performance in SVM, DT, and NV for six tests

One more crucial point in this experiment is decreeing the number of features by implementing preprocessing steps as shown in Fig. 3.

As depicted in Fig. 3, six different tests are implemented. It is clear that in RSNSW, the number of features is decreased significantly since the number of features in RS is 1638 and it becomes 1520 while in RSN it is 1553. Moreover, when RSNSW is tested,

0.8
0.6
F1 measure
0.4
0.2
0
RS RSN RSNSW RSNS RSNSWS RSNSSW
Six Expirements
SVM DT (C4.5) NB

Fig. 3. Number of features for six experiments

the number of features is 1634. On the other hand, the number of features is decreased by 37 and 16 for each RSNS and RSNSSW respectively to be compared with RS.

4 Conclusion and Future Work

In this study, detecting hate speech in Kurdish text is implemented by using three different classifiers on six various tests. The results are comparable for each test and different results are considerable. The F1 score is used for evaluating the results and comparing them. Based on the results, the best result F1 score for SVM, DT, and NV is 0.687, and 0.584 0.56 respectively. In the future, the preprocessing techniques can be improved and investigated to increase the high F1 score.

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